

MARION VANIER

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Applied ML + Embedded Systems + R&D Builder

PROFILE

Hands-on technical builder across machine learning, computer vision, embedded hardware, robotics and lab systems. 3+ years building field-facing ML/NDT systems for defense and industrial use, plus founder experience shipping a mobile computer-vision product. Strong fit for Boston startups, biomedical robotics teams, R&D labs, field systems groups and technical operations roles that need someone who can build, test and troubleshoot quickly.

CORE SKILLS

ML / CV: PyTorch, Hugging Face, diffusion models, transformers, computer vision, segmentation, anomaly detection, image rectification, synthetic data, signal processing, SSIM-based evaluation

Embedded / Hardware: NVIDIA Jetson, Linux, KiCad PCB design, high-voltage electronics up to 900 V, microcontrollers, soldering, BNC/coax cabling, CAD, 3D printing, safety procedures

Robotics / Software: Python, C++, C#, MATLAB, PID control, UAV systems, LiDAR simulation, MARSIM, FEA, Git, operator-facing tools, rapid prototyping

Domains: NDT, CT/tomography, sinogram reconstruction, acoustic emission analysis, edge AI, field inspection systems

EXPERIENCE

Antech Systems Inc. | Machine Learning & Embedded Systems Engineer | Chesapeake, VA | Jun 2022 - Present

- Built a diffusion-transformer pipeline for limited-angle CT/NDT sinogram enhancement. Raised reconstruction quality from about 0.30 baseline to about 0.85 using SSIM-informed evaluation and human visual review.
- Trained a rare-event crack-detection model on 1M+ inspection images with only 12 positive examples. Built data handling and evaluation around the class imbalance; work helped secure \$1M+ in U.S. Navy funding.
- Developed a foundation-style model for modal acoustic emission analysis from raw multi-channel sensor signals, moving NDT interpretation toward automated real-time analysis.
- Owned synthetic-data generation and curation for CT/NDT workflows. Found domain mismatch and low-variation failure modes, then converted reconstruction outputs into operator-readable defect maps.
- Designed and fabricated a high-voltage NDT multiplexer up to 900 V, including KiCad schematic design, component sourcing, assembly, custom coax cabling, enclosure design and safety procedures.
- Deployed ML inference and control logic on NVIDIA Jetson Linux edge hardware, integrating sensor data, model outputs, cabling and field-facing workflow.

Should I Slab This | Founder / Computer Vision Product Builder | Remote | 2025 - Present

- Built and launched a no-download mobile web MVP that analyzes trading-card centering from a phone photo and generates a shareable card passport. Early usage reached 100+ scans across 10 users.
- Built the image pipeline: segmentation, rectification, manual six-handle correction, centering ratios, scan correction flow and lightweight web deployment.
- Tested QR onboarding with card shops and shows, turning an ML demo into a product loop a non-technical collector can use in under a minute.

Mavericks One | UAV Systems & Software Consultant | Tampa, FL | May 2025 - Jan 2026

- Co-designed 7-inch UAV platforms for specialized payloads, including component sourcing, wiring, assembly, field flight testing and telemetry-driven iteration.
- Wrote custom C++ flight-control behavior and tuned PID controllers using real-world flight data.
- Set up LiDAR-based drone simulation workflows in MARSIM on Linux and built a C# video-analysis tool for AI scene segmentation and reporting.

Deaf LLC | Co-Founder & Systems Engineer | Remote | Jan 2018 - Jun 2022

- Co-founded a hardware startup developing a high-performance electric skateboard. Selected ESC, motor and battery systems, coordinated contract manufacturing and delivered a functional prototype under budget.

EDUCATION

Old Dominion University | Coursework toward B.S., Electrical & Computer Engineering | Norfolk, VA | 2018 - 2025
Computer Science minor coursework complete